

Ultrametric Root Counting

Martin Avendaño* and Ashraf Ibrahīm†

January 21, 2009

Abstract

If K is a complete non-archimedean field with a discrete valuation and $f \in K[X]$ is a polynomial with non-vanishing discriminant. The first main result of this paper is about connecting the number of roots of f to the number of roots of its reduction modulo a power of the maximal ideal of the valuation ring of K . If the polynomial f is regular, we give an algorithmic method to compute the exact number of roots of f in K and show that the number of roots of f equals the sum of the numbers of roots of its lower binomials.

1 Introduction

Let $f \in \mathbb{R}[X]$ be an univariate polynomial with real coefficients. Sturm's theorem allows us to determine the exact number of real roots of f in a given interval $[a, b]$. This is done by computing the difference between the number of sign changes of two sequences of real numbers called Sturm's sequences. We are interested in the analogue of Sturm's theorem over K , where K is a complete field with respect to non-archimedean discrete valuation. In this paper, we give an algorithmic method to compute the exact number of roots (total or with a given valuation) in K of a large class of polynomials in $K[X]$ called regular polynomials (see definition 2.2).

For any segment S of the Newton polygon of a regular polynomial $f \in K[X]$, the binomial containing the two terms of f corresponding to the vertices of S is called a lower binomial of f . We prove in theorem 4.6 that the number of roots of f in K^* coincides with the sum of the number of roots of all its lower binomials in K^* . Moreover, an explicit formula for the number of roots of each lower binomial is provided in theorem 4.5.

*Department of Mathematics, Texas A&M University, 3368 TAMU, College Station, Texas 77843-3368, USA. avendaño@math.tamu.edu, vov.math.tamu.edu/~avendaño
†Department of Mathematics, Texas A&M University, 3368 TAMU, College Station, Texas 77843-3368, USA. aibrahim@math.tamu.edu, vov.math.tamu.edu/~aibrahim

1

where $0 < i_1 < i_2 < \dots < i_{n_j+1} \leq n$. Note that in the case $k = 0$, the term $\delta = (-1)^{i_1} \alpha_{i_1} \alpha_{i_2} \dots \alpha_{i_n}$ appears in the sum (corresponding to a_{n-s_j}) and it has strictly minimum valuation. This means that $v(\delta) = v(a_{n-s_j}) = n \alpha_{i_1} + n_1 \alpha_{i_2} + \dots + n_{j-1} \alpha_{i_{n_j}}$. When $0 < k \leq n_j$ we can write

$$a_{n-s_j-k} = \delta b_{n_j-k} + \beta \quad (1)$$

where $\beta \in K$ collects all the terms appearing in a_{n-s_j-k} with $i_{n_j+1} > s_{j+1}$. This implies that $v(\beta) > n \alpha_{i_1} + n_1 \alpha_{i_2} + \dots + n_{j-1} \alpha_{i_{n_j}} + k \alpha_{i_{n_j+1}}$. Since f is a regular polynomial, we have that $v(a_{n-s_j-k}) > n \alpha_{i_1} + n_1 \alpha_{i_2} + \dots + n_{j-1} \alpha_{i_{n_j}} + k \alpha_{i_{n_j+1}}$ by the first item in the definition 2.2, and then $v(b_{n_j-k}) > k \alpha_{i_{n_j+1}}$. $\square$

3 Roots of the Reduced Polynomial

Consider a monic polynomial $f(X) = X^n + a_{n-1}X^{n-1} + \dots + a_1X + a_0 \in A[X]$. Assume that the discriminant $\Delta = \text{Res}_X(f, f')$ is non-zero and let $r = v(\Delta)$.

Lemma 3.1. For any $\alpha \in \overline{K}$ such that $f(\alpha) = 0$, we have $v(\alpha) \geq 0$.

Proof. Assume that $v(\alpha) < 0$. Since $f(\alpha) = 0$, we have that

$$\begin{aligned} n v(\alpha) &= v(\alpha^n) = v(a_{n-1} \alpha^{n-1} + \dots + a_0) \geq \min\{v(a_i \alpha^i) : 0 \leq i < n\} \\ &\geq \min\{v(\alpha^i) : 0 \leq i < n\} = (n-1)v(\alpha) \end{aligned}$$

and then $v(\alpha) \geq 0$. Contradiction. $\square$

The following lemma gives an estimation to the distance between the roots.

Lemma 3.2. If $f(X) = \prod_{i=1}^n (X - \alpha_i)$ with $\alpha_i \in \overline{K}$ for $i = 1, \dots, n$, then

$$v(\alpha_i - \alpha_j) \leq \frac{r}{2} \quad \forall i \neq j.$$

Proof. From the formula for the discriminant of a polynomial $\Delta = \prod_{1 \leq i < j \leq n} (\alpha_i - \alpha_j)^2$ we get $r = 2 \sum_{1 \leq i < j \leq n} v(\alpha_i - \alpha_j)$. Since all the roots satisfy $v(\alpha_i) \geq 0$, all the terms in this sum are non-negative. Therefore $v(\alpha_i - \alpha_j)$ can not exceed $r/2$ for any $i \neq j$. $\square$

We can derive from Lemma 3.2 an upper bound for the number of roots of f in K .

Corollary 3.3. The number of roots of f in K is $\leq q^{r/2n+1}$.

Proof. Otherwise we would have two roots $x, y \in A$ with $x \equiv y \pmod{p^{r/2n+1}}$, that is, $v(x - y) \geq r/2n + 1 > r/2$ in contradiction with Lemma 3.2. $\square$

On the other hand, Descartes' rule of signs implies that any univariate polynomial $f \in \mathbb{R}[X]$ with exactly $t \geq 1$ monomial terms has at most $2t - 1$ real roots. Note that, Descartes' bound over the reals doesn't depend on the degree of the polynomial and it is linear in the number of monomial terms. In [5], H.W. Lenstra gave an analogue of Descartes' bound over the p -adic numbers. He showed that if $f \in K[X]$ is an univariate polynomial with coefficients in K , a finite extension of the p -adic rationals $\mathbb{Q}_p$, with at most $t \geq 1$ monomial terms, then the number of roots of f in K is $O(t^2(q-1)\log(t))$ where q is the cardinality of the residue field of K . As a consequence of our result of root counting, we can improve Lenstra's bound to $O(t(q-1))$ for regular polynomials. All this work is done in sections 2 and 4.

In section 3 we obtain, in theorem 3.9 a reformulation of the classical Hensel's lifting. More precisely, we show a bijection between the set of roots of a monic polynomial (with coefficients in the valuation ring A of K) and some classes of roots of the reduced polynomial in $A/\pi^N A$ where π is the maximal ideal of A and N is a suitable positive integer. As a consequence, we prove that the number of roots in K of a polynomial in $K[X]$ with a given non-vanishing discriminant depends only on few "digits" of the coefficients (in corollary 3.10). It is easy to derive from the main result of this section an algorithm to compute the exact number of roots in K of a polynomial in $K[X]$.

2 Newton Polygon and Regularity

Let K be a complete field with respect to a non-archimedean discrete valuation v . We denote by $A = \{x \in K : v(x) \geq 0\}$ to the valuation ring of K , $\mathfrak{M} = \{x \in K : v(x) > 0\}$ to the maximal ideal of A , $\pi \in \mathfrak{M}$ to a generator of the ideal $\mathfrak{M}$ of A and $K_v = A/\pi A$ to the residue field of K with respect to v . We assume that K_v is finite with q elements and characteristic p and that $v(\pi) = 1$. We also denote by $\bar{K}$ the unique extension of the valuation of K to its algebraic closure $\bar{K}$.

Let $f(X) = a_n X^n + a_{n-1} X^{n-1} + \dots + a_1 X + a_0 \in K[X]$. The Newton polygon of f is the lower convex hull of the set of points

$$\{(i, v(a_i)) : i = 0, 1, \dots, n\}.$$

Theorem 2.1. Let $f(X) = a_n X^n + a_{n-1} X^{n-1} + \dots + a_1 X + a_0 \in K[X]$ such that $a_{n\pi} \neq 0$. Suppose S is the line segment in Newton polygon of f connecting the points $(i_s, v(a_{i_s}))$ and $(i', v(a_{i'}))$ with $s > s'$. Then f has exactly $s - s'$ roots with valuation m where $-m$ is the slope of the segment S . Moreover, f be can factorized as

$$f(X) = a_n \prod f_m(X)$$

where f_m is a monic polynomial in $K[X]$ with all roots of valuation m .

Let $f_N \in (A/\pi^N A)[X]$ denote the reduction of the polynomial f modulo π^N . We denote by $\beta_1, \dots, \beta_l \in A$ to the roots of f in K (by Lemma 3.1 we know that they are in A). It is clear that the reduction of any of these roots modulo π^N is a root of f_N . Unfortunately, the reduction modulo π^N does not give a bijection between the set of roots of f in K and the set of roots of f_N in $A/\pi^N A$ in general. However, we will show below that the reduction homomorphism is a bijection between the roots of f and clusters of roots of f_N defined by an equivalence relation. The inverse of the reduction homomorphism is given by a reformulation of the standard Hensel's Lemma.

In order to simplify the notation, we will denote $x \mapsto \overline{x}$ to the reduction modulo $\pi^N A$ homomorphism from A to $A/\pi^N A$.

Definition 3.4. Let $Z_N \subseteq A/\pi^N A$ be the set of roots of f_N . Two roots $x, y \in Z_N$ are in the same cluster (denoted by $x \approx y$) if and only if either $x = y$ and $N \leq r$ or $x \equiv y \pmod{\pi^{r+1}}$ and $N > r$. The cluster containing a root $x \in Z_N$ is written $[x]$. The set of clusters is written $Z_N/\approx$.

Lemma 3.5. If $N > r$ then the number of roots of f in K is $\leq |Z_N/\approx|$.

Proof. Write $f(X) = (X - \beta_1)(X - \beta_2) \dots (X - \beta_l)g(X)$ where g has no roots in K . Let $\beta_N = \overline{\beta} \in A/\pi^N A$ be the reduction of β , modulo $\pi^N A$. Since this reduction is a ring homomorphism, β_N is a root of f_N . Take $1 \leq i < j \leq l$. By lemma 3.2, we have $v(\beta_i - \beta_j) \leq r/2 \leq r$, i.e. $\beta_i \not\equiv \beta_j \pmod{\pi^{r+1}}$. Since $N > r$, we also have that $\overline{\beta_i} \not\equiv \overline{\beta_j} \pmod{\pi^{r+1}}$. This implies that $\beta_N \not\equiv \beta_N$ and then $[\beta_N] \neq [\beta_N]$. $\square$

Lemma 3.6. Let $\gamma \in A$ be such that $v(f(\gamma)) > r$. Then $v(f'(\gamma)) \leq r$.

Proof. Write $\Delta = a(X)f(X) + b(X)f'(X)$ with $a, b \in A[X]$ and evaluate at $X = \gamma$. Since $v(\Delta) \geq 0$, we have that $v(a(\gamma)f(\gamma)) > r$, and then $v(b(\gamma)f'(\gamma)) = v(\Delta - a(\gamma)f(\gamma)) = r$. We conclude that $v(b(\gamma)) \leq r$ because $v(b(\gamma)) \geq 0$. $\square$

In order to proceed, we need the following version of Hensel's Lemma.

Lemma 3.7 (Hensel). If $\gamma \in A$ satisfies $v(f(\gamma)/f'(\gamma)^2) > 0$ then there exists a root $\xi \in A$ of f such that $v(\xi - \gamma) \geq v(f(\gamma)/f'(\gamma))$.

Lemma 3.8. If $N > 2r$ then then number of roots of f in K is $\leq |Z_N/\approx|$.

Proof. Take $[\beta] \in Z_N/\approx$ and take some $\gamma \in A$ such that $\beta = \overline{\gamma}$. Since $f(\overline{\gamma}) = f_N(\gamma) = \overline{0}$, we have that $v(f(\gamma)) \geq N > 2r \geq r$. By Lemma 3.6 we have that $v(f'(\gamma)) \leq r$ and then $v(f(\gamma)/f'(\gamma)^2) > 0$. By Hensel's Lemma, there exists $\xi \in A$ such that $f(\xi) = 0$ and $\xi \equiv \gamma \pmod{\pi^{N-r}}$, because $v(f(\gamma)/f'(\gamma)) \geq N - r$. Since $N - r > r$ we have that $\xi \equiv \gamma \pmod{\pi^{r+1}}$ and also $\xi \equiv \overline{\gamma} \pmod{\pi^{r+1}}$ because $N > r$. This means that $[\beta] = [\xi]$. $\square$

Proof. See [4], Proposition 3.1.1. $\square$

Definition 2.2. A polynomial $f(X) = a_n X^n + a_{n-1} X^{n-1} + \dots + a_1 X + a_0 \in K[X]$ is regular if for any segment $S = (s, v(a_s)) \rightarrow (s', v(a_{s'}))$ of the Newton polygon (with $s > s'$), we have:

1. S does not contain any point from the set $\{(i, v(a_i)) : i = 1, \dots, n\}$ except the points $(s, v(a_s))$ and $(s', v(a_{s'}))$.
2. $p \mid s - s'$ where $p = \text{char}(K_v)$.

The polynomial $a_s X^s + a_{s'} X^{s'}$ is called the lower binomial of f corresponding to the segment S .

Theorem 2.3. Let $f(X) = X^n + a_{n-1} X^{n-1} + \dots + a_1 X + a_0 \in A[X]$ be a regular polynomial. Then all the factors $f_m(X)$ are also regular.

Proof. Let $\alpha_1, \alpha_2, \dots, \alpha_n \in \overline{K}$ be all the roots of f . Assume that

$$\begin{aligned} v(\alpha_1) &= \dots = v(\alpha_{n_1}) = m_1 \\ v(\alpha_{n_1+1}) &= \dots = v(\alpha_{n_2}) = m_2 \\ &\vdots \\ v(\alpha_{n_{j-1}+1}) &= \dots = v(\alpha_{n_j}) = m_{j+1} \\ &\vdots \\ v(\alpha_{n_{j+1}+1}) &= \dots = v(\alpha_{n_{k+1}}) = m_{k+1} \end{aligned}$$

where $m_1 < m_2 < \dots < m_{k+1}$. In order to keep a consistent notation we set $s_0 = 0$ and $s_{k+1} = n$. Let y be the factor $f_{m_{j+1}}$ of f and let $n_j = s_{j+1} - s_j$ be the degree of g .

$$\begin{aligned} g(X) &= (X - \alpha_{n_{j+1}})(X - \alpha_{n_{j+1}+1}) \dots (X - \alpha_{n_{k+1}}) \\ &= X^{n_j} + b_{n_j-1} X^{n_j-1} + \dots + b_1 X + b_0. \end{aligned}$$

The coefficient b_{n_j-k} , where $0 < k < n_j$, can be written as

$$b_{n_j-k} = (-1)^k \sum \alpha_{i_1} \alpha_{i_2} \dots \alpha_{i_k}$$

where the sum ranges over the sequences $s_j < i_1 < i_2 < \dots < i_k \leq s_{j+1}$. On the other hand,

$$a_{n-s_j-k} = (-1)^k \sum \alpha_{i_1} \alpha_{i_2} \dots \alpha_{i_{k+j}}$$

Note that if ξ and ξ' are two different roots of f in A , then $v(\xi - \xi') \leq r/2 \leq r$ by Lemma 3.2. This implies that $\xi \not\equiv \xi' \pmod{\pi^{r+1}}$, $\xi \not\equiv \xi' \pmod{\pi^N}$ and $[\xi] \neq [\xi']$. We conclude from here that the procedure described above gives a well defined map from the set $Z_N/\approx$ to the set of roots of f in K (we can not lift the same cluster to two different roots). Moreover, this map is injective, because we can reconstruct the cluster from the lifted root. $\square$

As an immediate consequence of Lemmas 3.5 and 3.8, we have the following theorem.

Theorem 3.9. For any $N > 2r$, the number of roots of f in K is equal to $|Z_N/\approx|$. More precisely, the map $x \mapsto [x]$ is a bijection between the set of roots of f in A (or in K) and $Z_N/\approx$.

Corollary 3.10. Let $g = X^n + b_{n-1} X^{n-1} + \dots + b_0 \in A[X]$ be such that $v(a_i - b_i) > 2r$. Then f and g have the same number of roots in K .

Proof. Since $a_i \equiv b_i \pmod{\pi^{2r+1}}$, then $\text{Res}_X(g, g') \equiv \text{Res}_X(f, f') \pmod{\pi^{2r+1}}$. Then the discriminant of g has also valuation r . We conclude by applying Theorem 3.9 to f and g with $N = 2r + 1$. $\square$

It is important to note here that the proofs of Lemmas 3.5 and 3.8 remain valid if we change our equivalence relation $\approx$ by the (apparently finer) relation $\sim$ defined by $x \sim y$ if and only if $x \equiv y \pmod{\pi^{N-r}}$. Therefore also Theorem 3.9 remains true with this new equivalence relation. Denote by $[[x]]$ to the cluster of roots with respect to $\sim$ that contains x . It is clear that $[[x]] \subseteq [x]$ for all x . On the other hand, the number of clusters with respect to $\sim$ or $\approx$ must be the same (they coincide with the number of roots of f in K), thus $[[x]] = [x]$ for all root $x \in A/\pi^N A$ of f_N . We derive several corollaries from this remark.

Corollary 3.11. For any $N > 2r$, the number of roots of f_N in $A/\pi^N A$ is less than or equal to p^r times the number of roots of f in K .

Proof. Any cluster $[[x]]$ contains at most p^r elements and the number of clusters is the number of roots of f in K . $\square$

Corollary 3.12. For any $N > 2r$, the number of roots of f_N is $A/\pi^N A$ is $\leq p^{r+1/2n+1}$.

Proof. Apply Corollaries 3.3 and 3.11. $\square$

Corollary 3.13. If $r = 0$ then the number of roots of f_N in $A/\pi^N A$ coincide with the number of roots of f in K for all $N \geq 1$.

Proof. Apply Corollary 3.11 and Lemma 3.5. $\square$

4 Roots of Regular Polynomials

The following corollary is just an special case of Theorem 3.9, when $r = 0$ and $N = 1$, but we are going to use it in this section, so we would like to state it as a separate result.

Corollary 4.1. *If $r = 0$ then the number of roots of f in K^* is equal to the number of roots of f_1 in K_r^* where f_1 is the reduction of f modulo πK .*

Lemma 4.2. *If $f(X) = X^n + a_0$ then*

$$\Delta(f) = (-1)^{n(n-1)/2} n! a_0^{n-1}.$$

Proof. Write $f(X) = X^n + a_0 = \prod_{i=1}^n (X - \alpha_i)$ with $\alpha_i \in \bar{K}$. Then

$$\begin{aligned} \Delta(f) &= (-1)^{n(n-1)/2} \text{Res}(f, f') = (-1)^{n(n-1)/2} \prod_{i=1}^n f'(\alpha_i) = (-1)^{n(n-1)/2} \prod_{i=1}^n n \alpha_i^{n-1} \\ &= (-1)^{n(n-1)/2} n^n \left(\prod_{i=1}^n \alpha_i \right)^{n-1} = (-1)^{n(n-1)/2} n^n (-1)^{n(n-1)} a_0^{n-1} \\ &= (-1)^{n(n-1)/2} n^n a_0^{n-1}. \end{aligned}$$

□

Lemma 4.3. *If $g(X) = X^n + a_{n-1}X^{n-1} + \dots + a_1X + a_0 \in A[X]$ satisfies $v(a_0) = 0$, $v(a_i) > 0$ for all $1 \leq i < n$ and $p \nmid n$ then the number of roots of g in K^* is equal to the number of roots of the lower binomial $X^n + a_0$ of g in K^* .*

Proof. By Lemma 4.2, the discriminant of $X^n + a_0$ has valuation 0. On the other hand, the polynomial g satisfies the hypothesis of Corollary 3.10 with respect to $f = X^n + a_0$. Then both f and its lower binomial have the same number of roots in K . □

Definition 4.4. *Let $a \in K^*$ be an element with valuation $v(a) = l$. The first digit of a is $\delta(a) = a/l^2 \in K_r^*$.*

The following result gives a procedure to count the number of roots of regular polynomial when its Newton polygon consists of only one line segment.

Theorem 4.5. *Let $f(X) = X^n + a_{n-1}X^{n-1} + \dots + a_1X + a_0 \in A[X]$ with $p \nmid n$ and $a_0 \neq 0$. Write $l = v(a_0)$ and assume that $v(a_{n-i}) > il/n$ for all $i = 1, \dots, n-1$. Then the number R of roots of f in K^* is equal to the number of roots of the lower binomial $X^n + a_0$ in K^* . Moreover, if $n \nmid l$ we have $R = 0$, and if $n \mid l$ then*

$$R = \begin{cases} \gcd(n, q-1) & \text{if } -\delta(a_0) \text{ is an } n\text{-th power in } K_r, \\ 0 & \text{otherwise.} \end{cases}$$

7

Proof. By Theorem 2.1, all the roots of both f and $\bar{f} = X^n + a_0$ have valuation $e = l/n$. It is clear that if $n \nmid l$, then neither f nor $\bar{f}$ have a root in K , because all the elements in K have integer valuation. Therefore, we only need to consider the case $n \mid l$.

Define $h(X) = \pi^{-l} f(\pi^l X)$. It is clear that f and h have the same number of roots in K . Our assumptions on the coefficients of f guarantee that h is a monic polynomial in $A[X]$. Moreover, if $h = X^n + b_{n-1}X^{n-1} + \dots + b_0$, then $v(b_0) = 0$ and $v(b_{n-i}) > 0$ for all $1 \leq i < n$. By Lemma 4.3, the number of roots of h in K coincides with the number of roots of its lower binomial $\bar{h} = X^n + \pi^{-l} a_0$ in K . Since $h(X) = \pi^{-l} f(\pi^l X)$, then $\bar{f}$ and $\bar{h}$ have the same number of roots in K . We conclude that f , $\bar{f}$, h and $\bar{h}$ have all the same number R of roots in K .

It only remains to prove the formula for R . By Lemma 4.2, the discriminant of $\bar{h}$ has valuation 0 (since $p \nmid n$ and $v(b_0) = 0$). Therefore, by Corollary 4.1, the number of roots R of $\bar{h}$ in K coincide with the number of roots in K_r of the reduction $\bar{h}_1 = X^n + \delta(a_0)$ of $\bar{h}$ modulo $\mathfrak{M}$. If $-\delta(a_0)$ is not an n -th power in K_r , then $\bar{h}$ has no roots. Otherwise, the number of roots of $\bar{h}$ in K_r coincide with the number of n -th roots of the unity in K_r . Since K_r^* is a cyclic group with $q - 1$ elements, $R = \gcd(q - 1, n)$ in this case. □

Theorem 4.6. *Let $f = a_n X^n + \dots + a_0 \in K[X]$ be a regular polynomial. Then the number of roots of f in K^* is equal to the sum of the number of roots of all its lower binomials in K^* .*

Proof. By theorem 2.1, we can factorize f as $f = a_0 \prod_{j=0}^r f_j$ where $f_0, \dots, f_r \in K[X]$ are monic polynomials and all the roots of each f_j have the same valuation m_{j+1} . Here $r+1$ is the number of segments of the Newton polygon of f and $-m_1 > \dots > -m_{r+1}$ are the slopes of these segments. Following the notation of theorem 2.3 we define $n_{j+1} = \deg(f_j)$ and $s_j = \{ \alpha \in \bar{K} : f(\alpha) = 0 \wedge v(\alpha) \leq m_j \}$. Setting $s_0 = \emptyset$ we have $n_j = s_{j+1} - s_j$. The lower binomials of f are the polynomials $g_j = a_{n-n_j} X^{n-n_j} + a_{n-n_{j-1}} X^{n-n_{j-1}}$. Let R and R_j denote the number of roots in K^* of f and f_j respectively. It is clear that $R = R_1 + \dots + R_r$. By theorem 2.3 the polynomials f_j are regular, and then, by theorem 4.5 its number R_j of roots in K^* depends only on its degree and the first digit of its independent coefficient. In order to conclude we only need to proof that R_j coincides with the number of roots of g_j in K^* . The number of roots of the lower binomial $g_j = a_{n-n_j} X^{n-n_{j+1}} (X^{n_{j+1}-n_j} + a_{n-n_{j+1}}/a_{n-n_j})$ in K^* coincide with the number of roots of the regular monic polynomial $X^{n_{j+1}-n_j} + a_{n-n_{j+1}}/a_{n-n_j}$ in K^* . The degree of this polynomial is $n_j = \deg(f_j)$ and by the equation 1 (with $k = n_j$) in the proof of theorem 2.3, the first digit of $a_{n-n_{j+1}}/a_{n-n_j}$ is equal to the first digit of the independent coefficient of f_j . Therefore R_j is also the number of roots of g_j in K^* . □

8

5 Conclusion

Since the regular polynomials are a large class of polynomials, we conjecture that our linear bound for the number of roots holds also for general polynomials. We are currently writting a second paper with a generalization of the notion of regularity to the multivariate case. Theorem 3.9 gives an algorithm to compute the exact number of roots (total or with a given valuation) of a polynomial $f \in \mathbb{Z}[X]$ in $\mathbb{Q}_p$. This is the p -adic analogue to Sturm's theorem.

Acknowledgments

The authors would like to thank J. Maurice Rojas for many fruitful discussions.

References

- [1] M. Ram Murty, *Introduction to p-adic Analytic Number Theory*, American Mathematical Society, 2002.
- [2] A. Robert, *A course in p-adic Analysis*, GTM, Vol.198, Springer-verlag, 2000.
- [3] W.H. Schikhof, *Ultrametric Calculus, An Introduction to p-adic Analysis*, Cambridge Studies in Adv.Math. 4, Cambridge Univ. Press, 1984.
- [4] E. Weiss, *Algebraic Number Theory*, McGraw-Hill, New York 1963.
- [5] H.W. Lenstra, *On the Factorization of Lacunary Polynomials*, Number Theory in Progress, vol. 1, Berlin 1999, pp. 277-291.

9